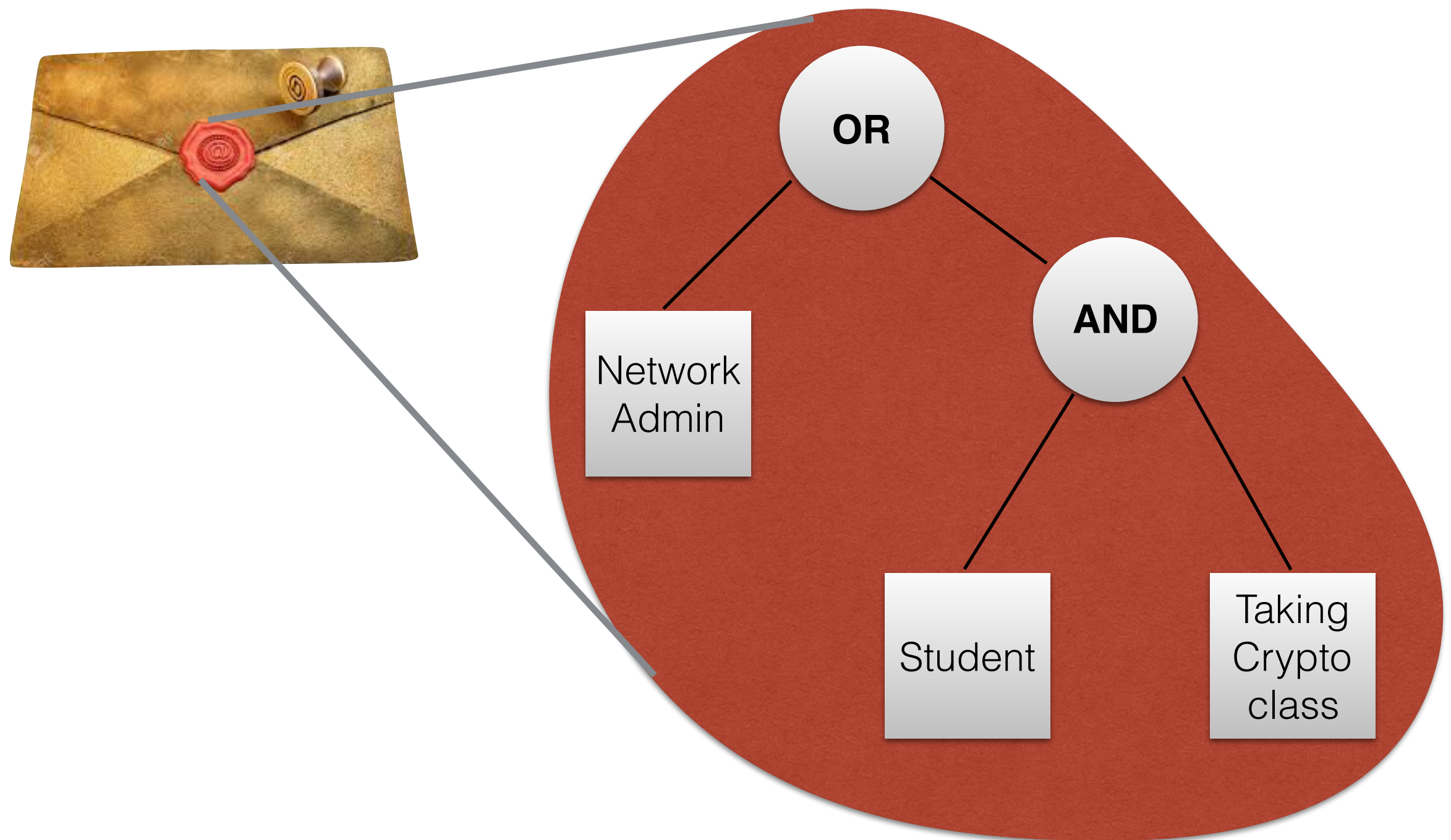


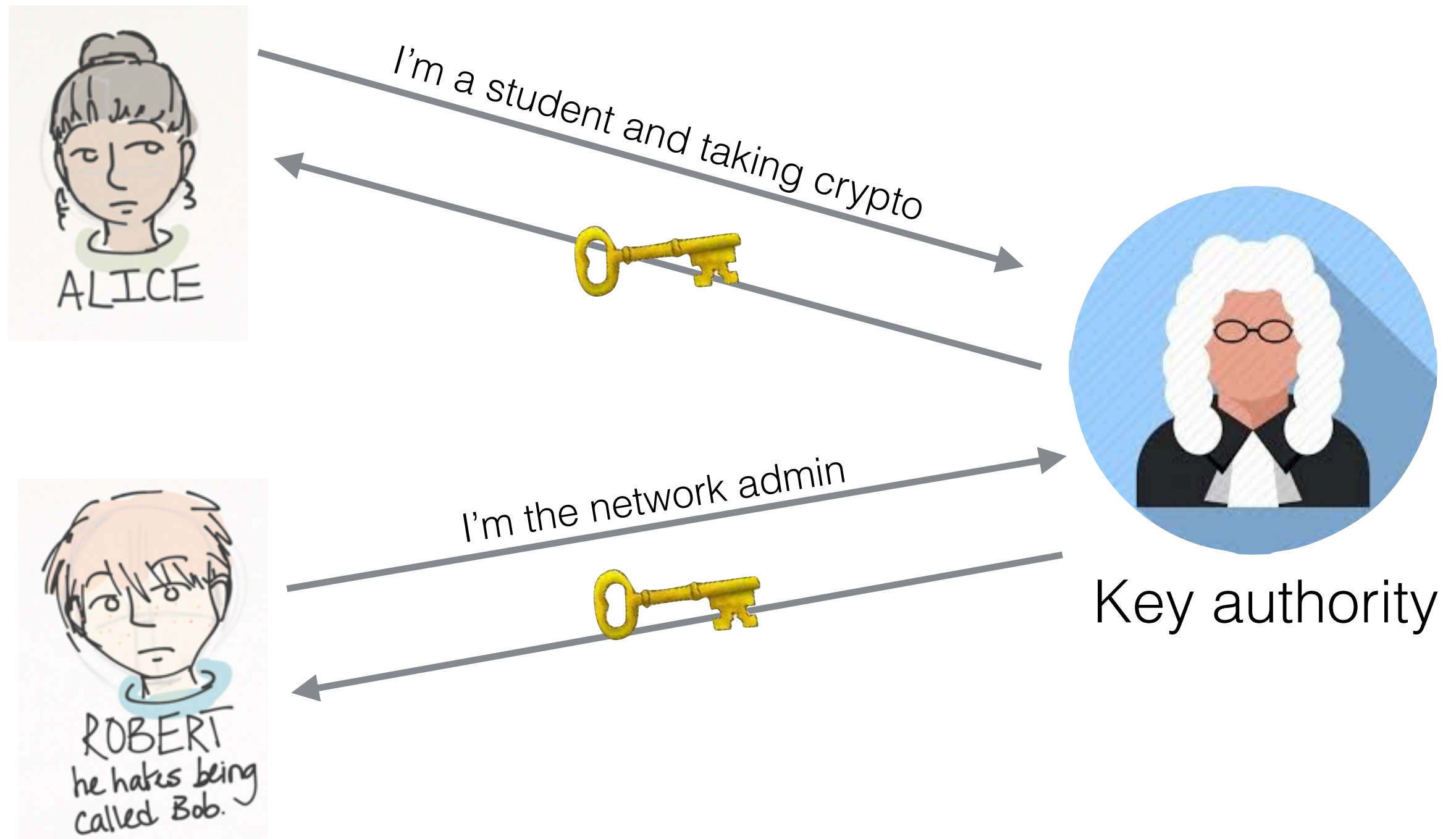
Decentralizing Policy- hiding Attribute-based Encryption

Yan Michalevsky, Marc Joye

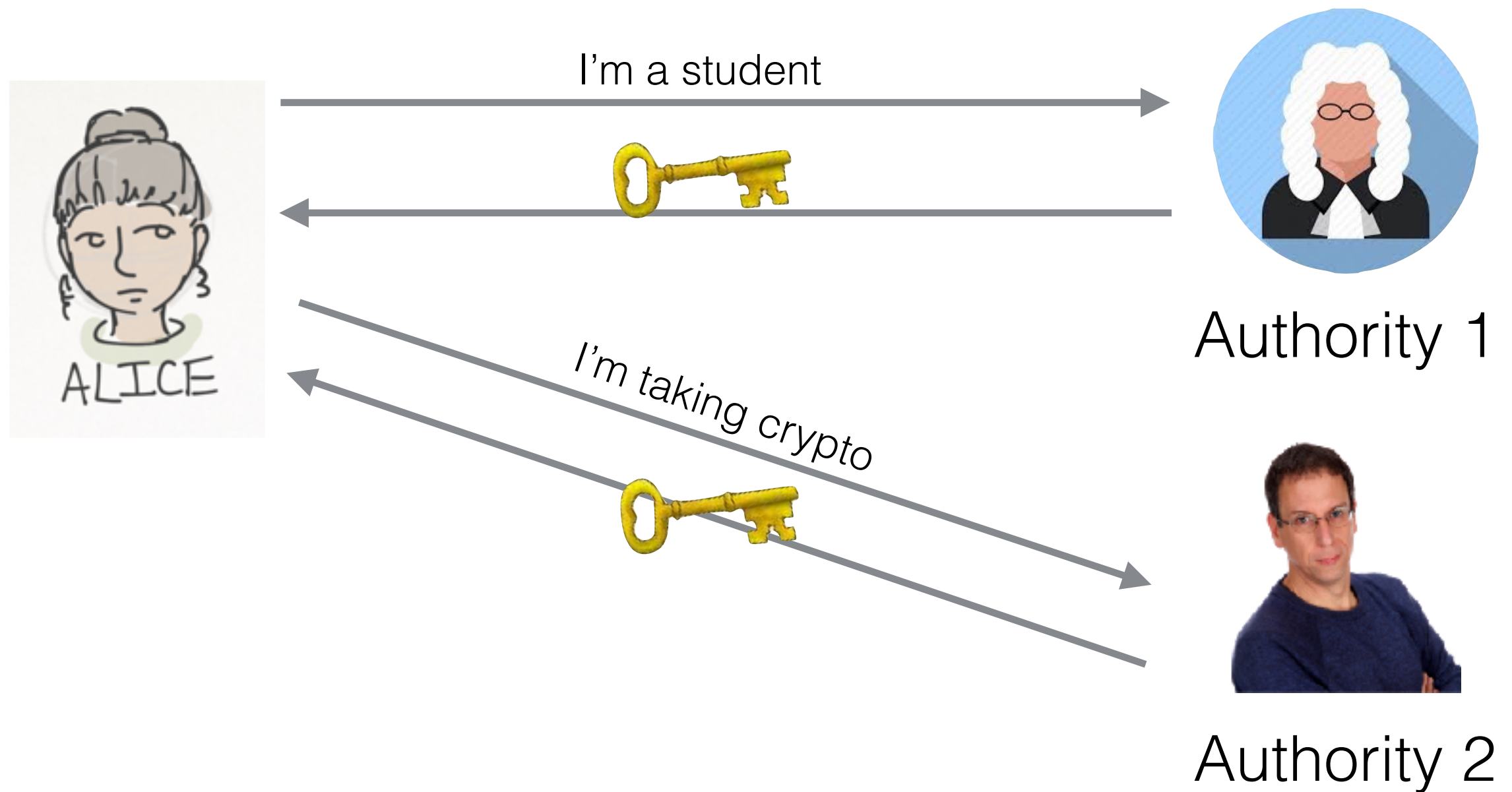
Attribute-based encryption (ABE)



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Multi-authority ABE



Collusion prevention



I'm a student



Authority 1



I'm taking crypto



Authority 2

Collusion prevention



I'm a student **GID=M**



Authority 1



I'm taking crypto **GID=E**



Authority 2

Decentralized ABE

- $Setup(\lambda)$: output public parameters PP
- $AuthSetup(PP, i)$: output authority public key PK_i
- $Encrypt(PP, \{PK_i\}, \pi, m)$: output ciphertext C
- $KeyGen(PP, \{PK_i\}, SK_i, \Sigma)$: output key $K_{i,GID}$
- $Decrypt(PP, \{K_{i,GID}\}, C)$: if keys satisfy policy, output m

Policy-hiding Predicate Encryption (PE)

- Someone can access, but we don't know what properties one has to have
- Zero Inner-Product PE (ZIPE):

- $Enc_{\bar{x}}(m) = c$

- $Dec_{SK_{\bar{v}}}(c) = \begin{cases} m : \langle \bar{x}, \bar{v} \rangle = 0 \\ \perp : \langle \bar{x}, \bar{v} \rangle \neq 0 \end{cases}$



Weak attribute-hiding

Adversary



Challenger



$$b \xleftarrow{R} \{0,1\}$$

$$PP, \{PK_i\}$$

key request $GID, \bar{v}$

$$K_{GID, \bar{v}}$$

$$m_0, m_1, \bar{x}_0, \bar{x}_1 \quad \langle \bar{v}, \bar{x} \rangle \neq 0$$

more key requests $\langle \bar{v}, \bar{x} \rangle \neq 0$

$$C \leftarrow \text{Encrypt}_{PP}(\bar{x}_b, m_b)$$

b'

What did we have?

- Decentralized ABE construction for Linear Secret Sharing [LW'11]
- Fully-hiding predicate encryption [KSW'08, KSW'13]
- Weakly attribute-hiding ABE [CGW'16, ...]
- Theoretic solution to Weakly Attribute-hiding Decentralized ABE [GKW'17, WZ'17]

We propose

- Efficient weakly attribute-hiding multi-authority ABE

ABE from ZIPE

Conjunctions: $A_1 \wedge A_2 \wedge A_4$

$$\overbrace{\begin{pmatrix} 1 & 1 & 0 & 1 & -3 \end{pmatrix}}^{\bar{x}^T} \bullet \overbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}}^{\bar{v}} = 0$$

ABE from ZIPE

Exact Threshold: 2-out-of-4

$$\overbrace{\begin{pmatrix} 1 & 1 & 1 & 1 & \underbrace{-2}_{-threshold} \end{pmatrix}}^{\bar{x}^T} \bullet \overbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}}^{\bar{v}} = 0$$

receiver has

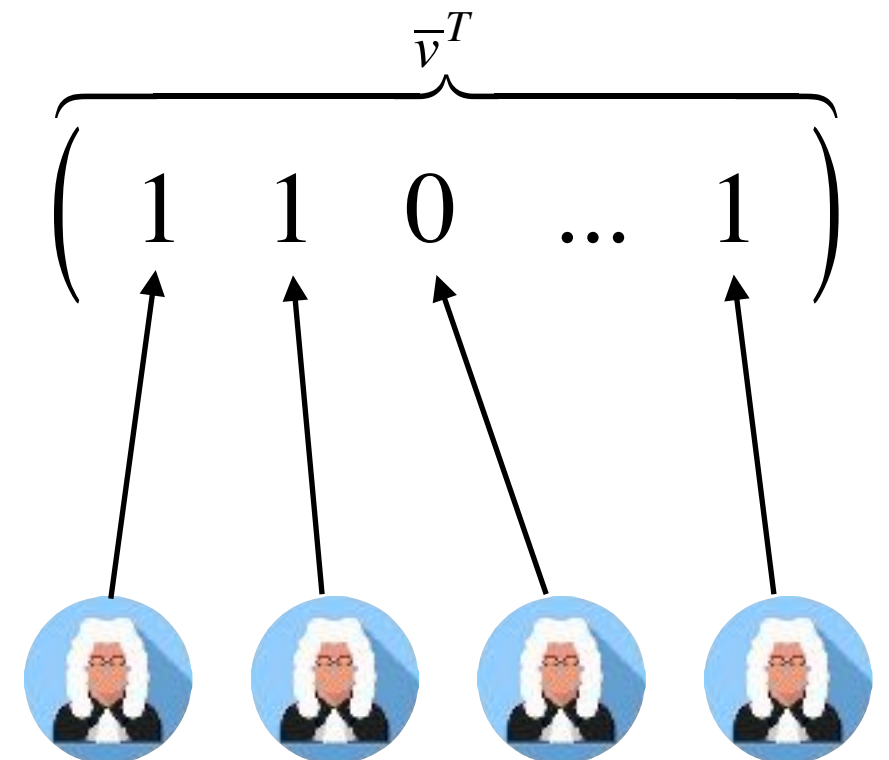
A_2, A_3

Our construction

- Based on decentralized ZIPE
- Based on the dual-system groups and predicate-encodings framework by Chen, Gay and Wee [CGW'17]
- Secure in the ROM, based on the k -linear assumption
- Requires a single-round coordination between authorities
 - Adding authorities/attributes requires to re-issue key-requests
- One authority needs to be honest (can be curious)
 - Can easily split this trust to multiple authorities

Decentralized ZIPE

- $Setup(\lambda)$: output public parameters PP
- $AuthSetup(PP, i)$: output authority pk and sk  _{i}  _{i}
- $Encrypt(PP, \{\text{green key icon}\}_i, \bar{x}, m)$
- $KeyGen(PP, \{\text{green key icon}\}_i, \text{red key icon}_i, \bar{v})$
- $Decrypt(PP, \{\text{yellow key icon}\}_{i, \text{GID}, \bar{v}}, C, \bar{v})$



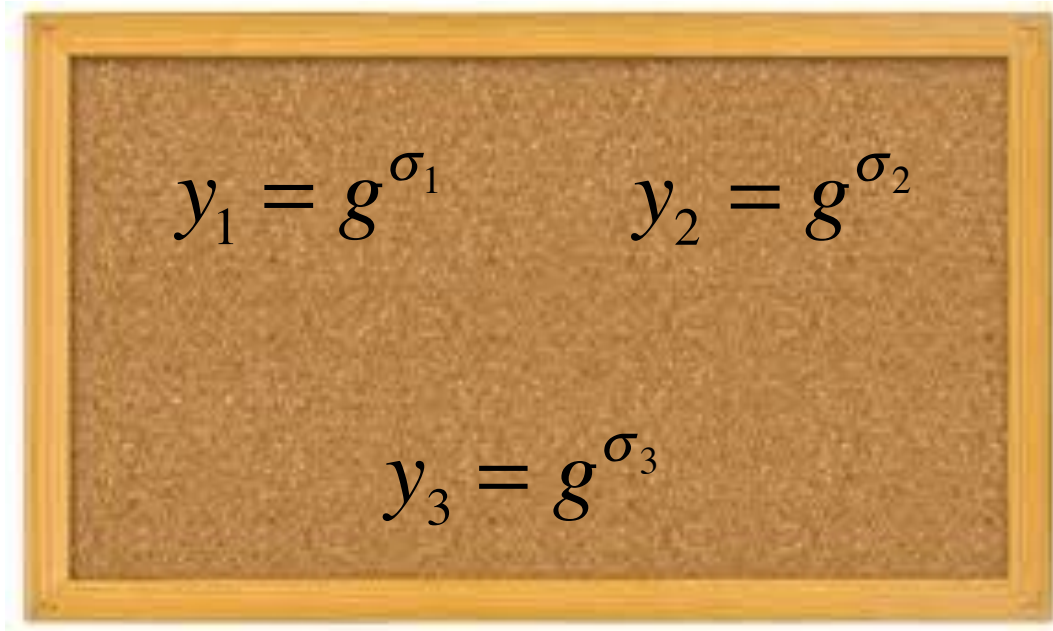
Authorities Setup



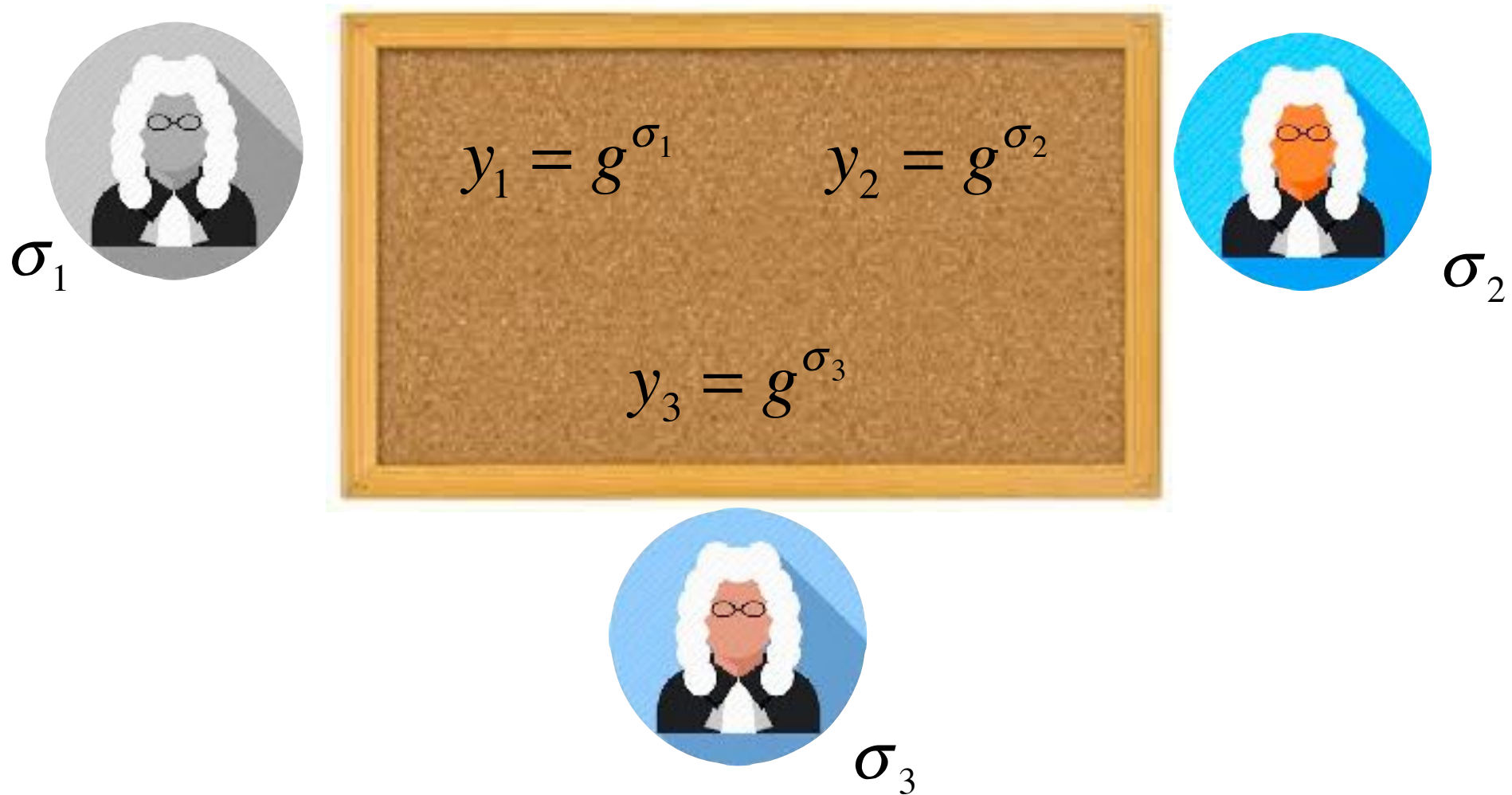
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Authorities Setup

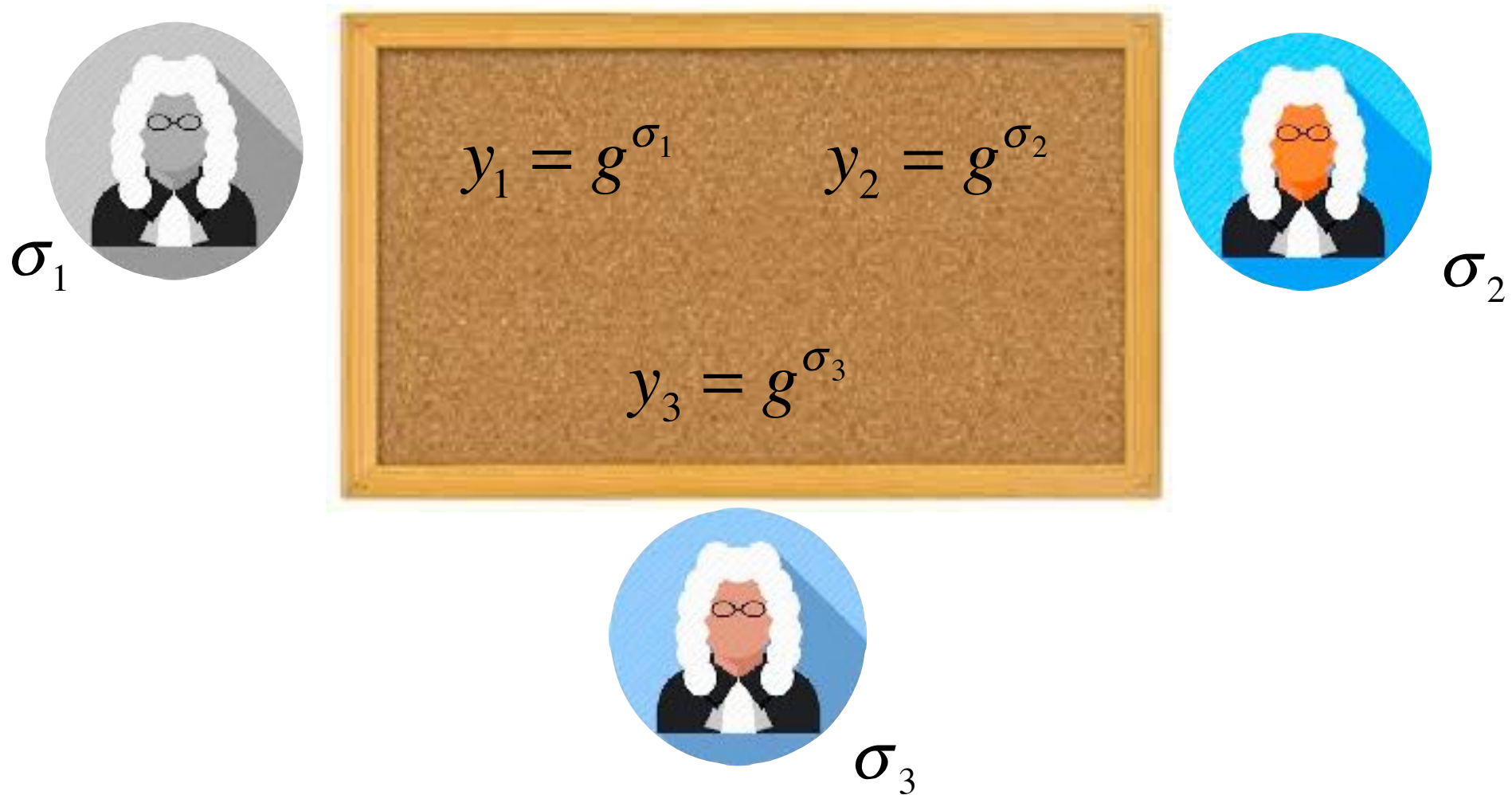


Authorities Setup



$$\mu_i = \sum_{j=1}^{i-1} H'(y_j^{\sigma_i}, \text{GID}, \bar{v}) - \sum_{j=i+1}^n H'(y_j^{\sigma_i}, \text{GID}, \bar{v})$$

Authorities Setup



$$\mu_i = \sum_{j=1}^{i-1} H'(y_j^{\sigma_i}, \text{GID}, \bar{v}) - \sum_{j=i+1}^n H'(y_j^{\sigma_i}, \text{GID}, \bar{v})$$

$$\sum_{i=1}^n \mu_i = 0$$

Key Generation

Key: $K_i = g_2^{\bar{\alpha}_i - v_i W_i \bar{h} + \mu_i}$

where $g_2^{\bar{h}} = H(\text{GID}, \bar{v})$

To decrypt, we combine the key-parts:

$$K = \prod_{i=1}^n K_i = g_2^{\sum_{i=1}^n \bar{\alpha}_i - v_i W_i \bar{h}}$$

resulting in cancellation of the masking terms

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Receiver Privacy

- *KeyGen* requires to provide the attribute vector $\bar{v}$
- Reveals the attributes of the user to all authorities
- Instead, we can commit to the vector, and selectively reveal certain elements to relevant authorities

$$H(GID, \bar{v}) \rightarrow H(GID, C) \quad C = Comm(\bar{v})$$

$$P_i = Open(C, i, v_i)$$

$$Verify(C, i, v_i, P_i)$$

Corrupt Authorities

- Can unmask their key share by revealing μ_i
 - Enables testing whether $x_i = 0$
- Can issue keys for arbitrary attributes v_i

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Exact Threshold: 2-out-of-4

Corrupt authority #2

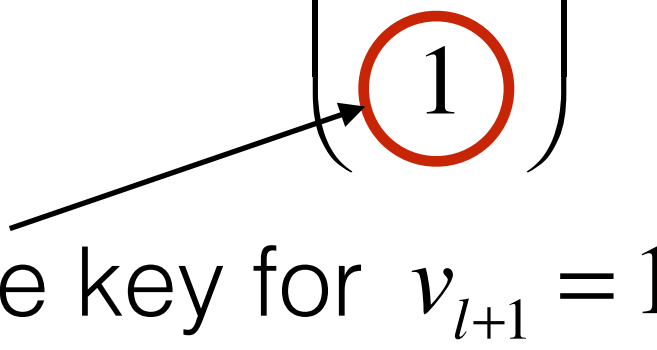
$$\overbrace{\begin{pmatrix} 1 & 1 & 1 & 1 & \underbrace{-2}_{-threshold} \end{pmatrix}}^{\bar{x}^T} \bullet \overbrace{\begin{pmatrix} 0 \\ 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}}^{\bar{v}} = 0$$

Immunizing against corrupt authorities

- We randomize the ciphertext policy vector
 - The adversary has negligible probability to find another attribute vector that solves $\langle \bar{v}, \bar{x} \rangle = 0$, except for the valid one
- Proposed encodings for:
 - Conjunctions
 - Hidden-vector encryption (HVE)
 - Threshold policies
 - CNF and DNF formulas

Secure encoding for conjunctions

$$A_1 \wedge A_2 \wedge A_4$$

$$\overbrace{\begin{pmatrix} r_1 & r_2 & 0 & r_4 & -(r_1 + r_2 + r_4) \end{pmatrix}}^{\bar{x}^T} \bullet \overbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}}^{\bar{v}} = 0$$


need to trust this one to issue key for $v_{l+1} = 1$

Secure threshold policy encoding

Threshold policy: 2-out-of-3

$$\begin{aligned}
 P(x) &= x^2 + a_1x + a_0 \\
 \forall i = 1..n : &\begin{cases} x_i = P(z_i) & i \in S \\ 0 & i \notin S \end{cases} \\
 x_{l+1} &= -P(0) = -a_0
 \end{aligned}
 \quad
 \overbrace{\left(\begin{array}{ccccc} P(z_1) & P(z_2) & 0 & P(z_4) & -a_0 \end{array} \right)}^{\bar{x}^T} \cdot \overbrace{\begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \\ \mathbf{1} \end{pmatrix}}^{\bar{v}} = 0$$

Receiver computes Lagrange coefficients and asks for corresponding keys

Secure encoding for CNF

Policy (CNF): $(A_1 \vee A_2) \wedge A_3$

$$P(x,y,z) = r_1(x-1)(y-1) + r_2(z-1) = r_1xy - r_1x - r_1y + r_2z + (r_1 - r_2)$$

$$\begin{array}{c}
 \overbrace{\left(\underbrace{-r_1}_x \quad \underbrace{-r_1}_y \quad \underbrace{r_2}_z \quad \underbrace{0}_{xyz} \quad \underbrace{r_1}_{xy} \quad \underbrace{0}_{xz} \quad \underbrace{0}_{yz} \quad \underbrace{r_1 - r_2}_{P(0,\dots,0)} \right)}^{\bar{x}^T} \bullet \begin{array}{c} \overbrace{\left(\begin{array}{c} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{array} \right)}^{\bar{v}} \\ \\ A_1 \wedge A_3 \end{array} = 0 \\
 \vdots \\
 \overbrace{\left(\underbrace{-r_1}_x \quad \underbrace{-r_1}_y \quad \underbrace{r_2}_z \quad \underbrace{0}_{xyz} \quad \underbrace{r_1}_{xy} \quad \underbrace{0}_{xz} \quad \underbrace{0}_{yz} \quad \underbrace{r_1 - r_2}_{P(0,\dots,0)} \right)}^{\bar{x}^T} \bullet \begin{array}{c} \overbrace{\left(\begin{array}{c} 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{array} \right)}^{\bar{v}} \\ \\ (A_1 \vee A_2) \wedge A_3 \end{array} = 0
 \end{array}$$

Open problems

- Efficient fully decentralized policy-hiding ABE (no μ -s)
- Fully (strong) hiding
- More expressive policies

